

Data Dictionary

A data dictionary was created to show the main classes, objects and variables used in the program.

It explains the data type of each item and what it is used for in the game.

Full Data Dictionary (Including Classes & Objects)

Name	Type	Description
Game	Class	Controls the main game logic, scoring, timer, and gameplay
Duck	Class	Handles duck movement, animation, and collision
Cracker	Class	Handles cracker display and interaction with the duck
Questions	Class	Manages loading and displaying questions, and checking answers
End	Class	Controls the game over screen, player name and saving scores
Welcome	Class	Controls the start screen and menu options

Game Variables

Name	Type	Description
score	Integer	Stores the player's score
game_time	Integer	Total time specified for the game
start_time	Integer	Stores when the game started
remaining_time	Integer	Time left in the countdown
game_over	Boolean	Checks if the game has ended
question_locked	Boolean	Prevents multiple answers at once

Duck Variables

Name	Type	Description
duck	Object	Instance of the Duck class
x, y	Integer	Position of the duck (co-ordinates)
is_moving	Boolean	Checks if the duck is moving
velocity_y	Integer	Controls vertical movement
gravity	Integer	Controls falling speed
jump_strength	Integer	Controls jump height
sprite	Object	Image of the duck on canvas

Cracker Variables

Name	Type	Description
cracker	Object	Instance of the Cracker class
cracker_x, cracker_y	Integer	Position of the cracker (co-ordinates)
sprite	Object / None	Stores cracker image on canvas
position	List	Stores cracker coordinates

Questions Variables

Name	Type	Description
questions	List	Stores all questions
current_question	Dictionary	Stores current question data
question_count	Integer	Tracks question number
choices	List	Stores answer options
correct_index	Integer	Correct answer from file
correct_button_index	Integer	Button with correct answer
answer_buttons	List	Stores answer buttons

Leaderboard Variables

Name	Type	Description
scores	List	Stores leaderboard entries
record	Tuple	Stores (name, time, score)
name	String	Player's name
time_taken	Integer	Time taken to finish game
FILE_NAME	String	Name of leaderboard file

End Screen Variables

Name	Type	Description
time	Integer	Stores time taken
GameClass	Object	Reference to game class
name_entry	Object	Input box for player's name
score_window	Object	Pop-out window
name_text	Object	Displays player's name

General UI Variables

Name	Type	Description
root	Object	Main application window
canvas	Object	Used to display graphics
label	Object	Label element
btn	Object	Button element